

INFLUENCE OF HIGH-INTENSITY INTERVAL TRAINING ON ADAPTATIONS IN WELL-TRAINED CYCLISTS

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ABSTRACT. Laursen, P.B., C.M. Shing, J.M. Peake, J.S. Coombes, and D.G. Jenkins. Influence of high-intensity interval training on adaptations in well-trained cyclists. *J. Strength Cond. Res.* 19(3):527–533. 2005.—The purpose of the present study was to examine the influence of 3 different high-intensity interval training regimens on the first and second ventilatory thresholds (VT₁ and VT₂), anaerobic capacity (ANC), and plasma volume (PV) in well-trained endurance cyclists. Before and after 2 and 4 weeks of training, 38 well-trained cyclists ($\dot{V}O_{2peak} = 64.5 \pm 5.2 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) performed (a) a progressive cycle test to measure $\dot{V}O_{2peak}$, peak power output (PPO), VT₁, and VT₂; (b) a time to exhaustion test (T_{max}) at their $\dot{V}O_{2peak}$ power output (P_{max}); and (c) a 40-km time-trial (TT₄₀). Subjects were assigned to 1 of 4 training groups (group 1: $n = 8$, $8 \times 60\%$ T_{max} at P_{max}, 1:2 work-recovery ratio; group 2: $n = 9$, $8 \times 60\%$ T_{max} at P_{max}, recovery at 65% maximum heart rate; group 3: $n = 10$, 12×30 seconds at 175% PPO, 4.5-minute recovery; control group: $n = 11$). The TT₄₀ performance, $\dot{V}O_{2peak}$, VT₁, VT₂, and ANC were all significantly increased in groups 1, 2, and 3 ($p < 0.05$) but not in the control group. However, PV did not change in response to the 4-week training program. Changes in TT₄₀ performance were modestly related to the changes in $\dot{V}O_{2peak}$, VT₁, VT₂, and ANC ($r = 0.41, 0.34, 0.42$, and 0.40 , respectively; all $p < 0.05$). In conclusion, the improvements in TT₄₀ performance were related to significant increases in $\dot{V}O_{2peak}$, VT₁, VT₂, and ANC but were not accompanied by significant changes in PV. Thus, peripheral adaptations rather than central adaptations are likely responsible for the improved performances witnessed in well-trained endurance athletes following various forms of high-intensity interval training programs.

KEY WORDS. ventilatory threshold, anaerobic capacity, plasma volume, endurance performance, short-term training

INTRODUCTION

Considerable information is available concerning the physiological adaptations responsible for the improvements in endurance performance observed following high-intensity interval training in sedentary and recreationally trained individuals (13). In contrast, relatively little is understood concerning the mechanisms responsible for the performance improvements that occur following high-intensity interval training in the already well-trained endurance athlete (9, 10, 16). The limited research that has examined physiological adaptations in well-trained athletes to high-intensity interval training has revealed no change in oxidative or glycolytic enzyme activity, despite significant improvements in endurance performance (34). Instead, an increase in skeletal muscle buffering capacity (34), improvements in the ventilatory threshold (VT) (1, 12, 15), and enhanced fat oxidation (33) have been suggested as potential mechanisms responsible for the improvement in endurance

performance that is generally witnessed. Further potential physiological adaptations include an increase in plasma volume (PV) and anaerobic capacity (ANC).

Increases in PV with endurance training facilitate improvement in the delivery of oxygen to working muscles and reduce cardiovascular strain (27). These adaptations have been reported to increase both $\dot{V}O_{2max}$ and exercise time to fatigue by 4 and 11%, respectively, in untrained individuals (3). However, well-trained athletes already have a high PV (32), and although PV can be artificially increased in these individuals (31), whether this plays a role in improving stroke volume, $\dot{V}O_{2max}$, and/or performance remains equivocal (26, 31, 32). Although there have been reports of acute PV expansion (12–13%) in well-trained cyclists measured within the 24-hour period following an ultradistance cycling event (14, 25), it is not clear whether this transient expansion persists and/or whether it is related to the improved endurance performance that is often observed following high-intensity interval training in well-trained athletes.

As measured by the maximal accumulated oxygen deficit method (22), ANC is believed to be an important contributor to the performance of well-trained endurance athletes (2). Tabata et al. (30) have shown that 6 weeks of repeated supramaximal high-intensity interval training (170% $\dot{V}O_{2max}$) improved both $\dot{V}O_{2max}$ (+15%) and ANC (+28%) ($p < 0.05$) in 7 moderately trained subjects ($\dot{V}O_{2max} = \sim 50 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$). However, the response of ANC in well-trained athletes to different high-intensity interval training programs has, to our knowledge, not been examined. Moreover, although we have shown that the VT can be significantly increased following just 4 high-intensity interval training sessions during 2 weeks in cyclists (15), the response of the VT to a longer duration high-intensity interval training program and to different high-intensity interval training programs is yet to be described.

Because of the paucity of data that relate directly to our understanding of the physiological adaptations that occur following high-intensity interval training in the well-trained athlete, this study investigated adaptations to PV, ANC, and the first and second VTs (VT₁ and VT₂) that occurred in well-trained cyclists grouped into 3 separate training programs (19). Also reported in this article are the $\dot{V}O_{2}$, heart rate (HR), and blood lactate (Lac⁻) responses to the 40-km time trial (TT₄₀) measured throughout the training program (19). We hypothesized that high-intensity interval training prescribed using exercise intensities equivalent to the $\dot{V}O_{2max}$ cycling power output (P_{max}) with durations equivalent to 60% of the time to ex-

haustion ($T_{\max}$) at $P_{\max}$ (groups 1 and 2 in the present study) would lead to significant enhancements in PV, $\dot{V}O_{2\max}$, VT_1 , and VT_2 , whereas these variables would be not be altered in a group performing a high-intensity interval training program prescribed using repeated supramaximal sprinting (group 3). Instead, we hypothesized that ANC would be significantly enhanced in the repeated supramaximal sprinting group (group 3) and that performance in all high-intensity interval training groups would lead to significant improvements in TT_{40} performance.

METHODS

Experimental Approach to the Problem

This study used a repeated-measures design to examine the influence of prolonged high-intensity interval training bouts (groups 1 and 2) and repeated supramaximal sprinting (group 3) (8 sessions completed during 4 weeks) to determine whether possible improvements in TT_{40} performance in already well-trained cyclists were related more to adaptations in central aerobic variables (i.e., PV), peripheral aerobic variables (i.e., $\dot{V}O_{2\text{peak}}$, VT) or anaerobic related variables (i.e., ANC, Lac^-). The 4-week time frame of the high-intensity interval training intervention has been used previously to show changes in physiological variables in already well-trained cyclists (33, 34).

Subjects

Forty-one well-trained male cyclists and triathletes (mean $\pm$ SD: age = 25 ± 6 years; height = 180 ± 5 cm; mass = 75 ± 7 kg; $\dot{V}O_{2\text{peak}}$ = 64.5 ± 5.2 ml·kg⁻¹·min⁻¹) with a minimum of 3 years of cycle training experience (6 ± 3 years) volunteered for this study. Cycle training distance during the study was 285 ± 95 km·wk⁻¹, completed predominantly at a low intensity (<80% maximum HR), and subjects reported not to be engaged in a strength training program. After being fully informed of the risks and stresses associated with the study, subjects completed a medical history questionnaire and gave their written informed consent to participate in the study, which was approved by the Medical Research Ethics Committee of The University of Queensland.

Preliminary Testing

Preliminary testing was conducted during the off-season and precompetitive phase of the athletes' yearly training program. For 3 consecutive weeks before the training intervention, athletes reported to the laboratory 3 times each week to perform (a) a progressive exercise test to determine $\dot{V}O_{2\text{peak}}$ and peak power output (PPO), (b) a $T_{\max}$ test at the $P_{\max}$, and (c) a TT_{40} on their own road bicycle mounted to a stationary wind-trainer. Athletes reported to the laboratory for testing on alternate days in consecutive order, having not trained for at least 12 hours; the order of the tests remained the same on each testing occasion.

Exercise during the first week served to familiarize the subjects with the tests and training sessions, whereas the test results obtained during weeks 2 and 3 were used to calculate the coefficients of variation for the baseline measures. Subjects reported to a controlled environmental laboratory condition ($\sim 21^\circ\text{C}$, 40–60% relative humidity, 760–770 mm Hg) at the same time of day for all tests. During each test and on all occasions, incremental power

output, speed, and/or exercise time were blinded to the athlete. Athletes were asked to keep their eating habits constant before all tests and to avoid consuming food within 2 hours of exercise.

Progressive Exercise Test

The $\dot{V}O_{2\text{peak}}$ was determined using an electronically braked cycle ergometer (Lode Excalibur Sport; Lode Medical Technology, Groningen, The Netherlands) modified with clip-in pedals and low-profile racing handlebars. The saddle and handlebar positions of the cycle ergometer were adjusted to resemble each athlete's own bicycle, and subjects warmed up at a self-selected pace for 5 minutes. The incremental test commenced at an initial workload of 100 W and increased thereafter by 15 W·30 s⁻¹ until volitional fatigue. Expired air was analyzed for fractions of expired oxygen and carbon dioxide ($F_{\text{E}}O_2$ and $F_{\text{E}}CO_2$) every 30 seconds during exercise (Ametek gas analyzers SOV S-3A11 and COV CD3A, Ametek, Pittsburgh, PA), whereas minute ventilation ($\dot{V}_E$) was recorded every 30 seconds using a turbine ventilometer (model 096, Morgan, Kent, England). The gas analyzers were calibrated immediately before and validated after each test using a certified beta gas mixture (Commonwealth Industrial Gas Ltd., Brisbane, Australia); the ventilometer was calibrated before and validated after each test using a 1-L syringe in accordance with the manufacturer's instructions. The metabolic system was verified by the Australian Laboratory Standards Assistance Scheme (8). $\dot{V}O_{2\text{peak}}$ was recorded as the highest $\dot{V}O_2$ reading averaged over 2 consecutive readings, and the PPO was recorded as the highest 30-second power output completed during the incremental test. $\dot{V}O_{2\text{peak}}$ was defined by the following criteria: (a) the oxygen consumption ceased to increase linearly with a rising workload and approached a plateau or dropped slightly, the last 2 values agreeing within ± 2 ml·kg⁻¹·min⁻¹; (b) 90% of age-predicted peak HR was attained; and (c) respiratory exchange ratio (RER) was greater than 1.10. Both VT_1 and VT_2 were measured by 2 independent reviewers according to methods that have been recently described (20) and completed in our laboratory (15). Reviewers were blinded to each other's findings, and in the event of a disagreement, the opinion of a third investigator was sought.

$T_{\max}$ at $P_{\max}$

The $P_{\max}$ was calculated from the progressive exercise test and defined as the minimal power output that elicited a $\dot{V}O_2$ reading that was within 2 ml·kg⁻¹·min⁻¹ of the previous reading, despite an increase in workload. After a 5-minute warm-up at 100–250 W, subjects cycled to fatigue at $P_{\max}$ at a self-selected cadence; the test was stopped when cadence fell below 60 rev·min⁻¹. Both HR and $\dot{V}O_2$ were recorded at 20-second intervals throughout exercise. Athletes were blinded to the time elapsed on all testing occasions. This test has been recently reported to have a coefficient of variation of $6 \pm 6\%$ (18).

Anaerobic Capacity

The ANC was estimated by a modified version (4) of the maximal accumulated oxygen deficit method (22), using the $\dot{V}O_2$ averaged each 60 seconds from 100–280 W during the progressive exercise test to form a regression equation. Oxygen demand at $P_{\max}$ was derived using the relationship and solving for $\dot{V}O_2$ at $P_{\max}$, according to a pre-

viously published method (11). The oxygen demand (in $\text{ml}\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$) was multiplied by T_{max} (in seconds) to estimate the overall oxygen demand. Total accumulated $\dot{V}\text{O}_2$ (in $\text{ml}\cdot\text{kg}^{-1}$) during the T_{max} test was subtracted from the overall oxygen demand to determine the oxygen deficit (in $\text{ml}\cdot\text{kg}^{-1}\cdot\text{s}^{-1}$). No correction for use of oxygen stores was made. Oxygen deficit was expressed as $\text{ml}\cdot\text{min}^{-1}$.

Laboratory-Simulated 40-km Time Trial

A laboratory-simulated TT_{40} was completed on the athlete's own road bicycle mounted to a stationary wind-trainer (Cateye-Cyclosimulator CS-1000; Cateye Co. Ltd., Osaka, Japan). The rear tire was inflated to 120 lb/in_2 and placed gently against the friction device before securing; the spring-loaded release brake was removed, placing a wind-regulated friction load against the rear wheel. The athlete's same rear wheel was used for each TT_{40} . Total time to complete the 40-km trial was recorded for the calculation of average speed. Subjects were permitted to consume water ad libitum during the TT_{40} . This test has been recently reported to be reproducible within a coefficient of variation of $1 \pm 1\%$ (17). During the TT_{40} , HR was recorded at 2-minute increments, capillary fingertip samples of Lac^- (Accusport, Boehringer Mannheim, Mannheim, Germany) were analyzed at 10 and 30 km, and 3-minute samples of expired gases were collected at 5, 20, and 35 km.

High-Intensity Interval Training Protocols

Subjects were assigned and matched to create homogeneous groups based first on their TT_{40} performance and second on their $\dot{V}\text{O}_{2\text{peak}}$. Multisport athletes and cyclists were equally distributed throughout the high-intensity interval training groups. All high-intensity interval training groups trained twice per week for 4 weeks and were reassessed after 2 and 4 weeks of high-intensity interval training (19). The high-intensity interval training program workload was adjusted following the midtraining assessment. At each interval training session, cyclists in group 1 completed 8 intervals at P_{max} for a duration equal to $60\% T_{\text{max}}$, with a 1:2 recovery ratio. Group 2 performed the same work intervals as group 1, except that recovery time was based on HR returning to 65% of the maximum HR. Group 3 completed twelve 30-second bouts per session at 175% of PPO, separated by 4.5 minutes of recovery. The control group was reassessed at the same times, and subjects in this group were asked to maintain their regular low-intensity base-training program.

Blood Sampling

Blood was sampled before each progressive exercise test and the TT_{40} , and values following analysis were averaged for the week. Subjects sat quietly in a chair for 5 minutes before capillary fingertip blood samples were collected. Samples were collected in 50- μl heparinized capillary tubes for the determination of hematocrit (Hct), hemoglobin (Hb), and total plasma protein content (PRO). The Hct determinations were made in duplicate by centrifugation for 5 minutes using a microcapillary centrifuge (Jouan Centrifuge; Thermo Electron Co., San Jose, CA). Ten microliters of blood drawn in duplicate was stored in Drabkin's reagent for analysis of Hb colorimetrically (545 nm) after conversion to cyanmethemoglobin, and PV was calculated according to the methods of Dill and Costill (5). Ten microliters of plasma was placed in 1

ml of Biuret reagent and analyzed colorimetrically (540 nm). The PRO was calculated against a commercially prepared human serum standard (Randox Laboratories Ltd., Antrim, UK).

Statistical Analyses

Data from 2 athletes from group 1 and 1 athlete from group 2 were eliminated from the data analysis due to illness or failure to comply with the training program, leaving 8 subjects in group 1, 9 subjects in group 2, 10 subjects in group 3, and 11 subjects in the control group for the final analysis. A repeated-measures analysis of variance (ANOVA) was run separately in each specific training group to determine whether each particular high-intensity interval training program had an effect on the dependent measures. As well, a 4×3 (group $\times$ time) repeated-measures ANOVA compared changes in the dependent measures over time between groups. Dunnett's post hoc comparisons were used to determine whether the high-intensity interval training groups improved significantly more than the control group, whereas Tukey's post hoc test compared differences between high-intensity interval training groups. Pearson's product moment was used to examine relationships between variables. All statistics were run on SPSS statistical software, version 10.0 for Windows (SPSS Inc, Chicago, IL), with α set at 0.05. All data throughout are expressed as mean $\pm$ SD.

RESULTS

The influence of the different high-intensity interval training programs used in this study on various performance markers has been previously described (19). In brief, all high-intensity interval training groups improved their TT_{40} performance (+4.4 to +5.8%) and PPO (+3.0 to +6.2%) significantly more than the control group (-0.9 to $+1.1\%$; $p < 0.05$). However, only group 1 (+5.4%) and group 2 (+8.1%) improved their $\dot{V}\text{O}_{2\text{peak}}$ significantly more than the control group ($+1.0\%$; $p < 0.05$) (19). The coefficient of variation calculated from the preliminary tests on the dependent measures of Hct, Hb, PRO, ANC, VT_1 , and VT_2 were 1.0, 2.2, 1.9, 7.9, 5.4, and 3.1%, respectively.

Markers of PV

Table 1 gives the PV variables of PRO, Hb, Hct, and calculated ΔPV . No significant differences in these measures were found over time or among training groups.

VT_1 , VT_2 , and ANC

Mean values for VT_1 , VT_2 , and ANC measured throughout the high-intensity interval training programs are given in Table 2. VT_1 and VT_2 were significantly increased in all training groups ($p < 0.01$) but not in the control group. Moreover, the change in VT_1 was significantly greater in group 2 and group 3 compared with the control group (i.e., significant interaction; $p < 0.05$) but not in group 1 ($p = 0.06$). Although a statistical trend ($p < 0.2$) suggested that VT_2 may have changed more in group 1 and group 2 compared with the control group, changes in VT_2 were not significantly different between the groups. The ANC significantly increased over time in group 1 ($p < 0.05$), group 2 ($p = 0.05$), and group 3 ($p < 0.01$), but not in the control group. The change in ANC was not significantly different among the groups.

TABLE 1. Hematological measures of plasma volume measured throughout the training program.*

Measure	Before	During	After
Group 1			
PRO (g·L ⁻¹)	75.6 ± 4.5	73.8 ± 4.9	74.6 ± 3.0
Hct (%)	43.6 ± 2.2	44.8 ± 2.9	44.6 ± 1.8
Hb (g·dL ⁻¹)	14.2 ± 0.9	14.4 ± 1.2	14.4 ± 1.2
ΔPV (%)	0.0 ± 0.0	1.3 ± 5.8	1.3 ± 9.0
Group 2			
PRO (g·L ⁻¹)	78.4 ± 3.7	77.5 ± 3.5	75.7 ± 3.2
Hct (%)	43.7 ± 2.5	43.6 ± 1.6	43.3 ± 2.4
Hb (g·dL ⁻¹)	14.1 ± 1.3	14.1 ± 1.1	14.0 ± 1.4
ΔPV (%)	0.0 ± 0.0	0.1 ± 2.9	0.3 ± 5.1
Group 3			
PRO (g·L ⁻¹)	77.2 ± 2.9	76.2 ± 3.5	75.2 ± 4.6
Hct (%)	43.1 ± 1.6	43.3 ± 1.9	43.0 ± 1.6
Hb (g·dL ⁻¹)	14.2 ± 0.9	14.2 ± 0.8	14.0 ± 1.9
ΔPV (%)	0.0 ± 0.0	0.1 ± 5.1	1.7 ± 8.6
Control group			
PRO (g·L ⁻¹)	77.3 ± 2.7	77.1 ± 5.1	77.4 ± 4.5
Hct (%)	44.1 ± 2.3	44.1 ± 2.4	44.2 ± 1.9
Hb (g·dL ⁻¹)	14.7 ± 1.1	14.8 ± 1.6	14.7 ± 1.3
ΔPV (%)	0.0 ± 0.0	-0.7 ± 4.2	0.0 ± 4.9

* No significant changes were found over time among the groups. PRO = plasma protein content; Hct = hematocrit; Hb = hemoglobin; ΔPV = calculated change in plasma volume.

Physiological Variables During TT₄₀ Performance

The TT₄₀ performance, HR, Lac⁻, $\dot{V}O_2$, and RER measured during the high-intensity interval training program are presented in Table 3. Mean HR during the TT₄₀ performance significantly increased throughout the high-intensity interval training program in all high-intensity inter-

TABLE 3. Changes in the mean heart rate (HR), blood lactate (Lac⁻), oxygen consumption ($\dot{V}O_2$), and respiratory exchange ratio (RER) during the 40-km time trial throughout the high-intensity interval training program.

Measure	Before	During	After
Group 1			
HR (bpm)	172 ± 14	174 ± 14	178 ± 12*
Lac ⁻ (mmol·L ⁻¹)	7.1 ± 2.4	7.5 ± 1.9	9.3 ± 2.8*
$\dot{V}O_2$ (L·min ⁻¹)	4.30 ± 0.42	4.34 ± 0.45	4.45 ± 0.40*
RER	1.02 ± 0.04	1.02 ± 0.04	1.03 ± 0.03
Group 2			
HR (bpm)	173 ± 11	174 ± 5	179 ± 7*
Lac ⁻ (mmol·L ⁻¹)	7.1 ± 2.5	7.2 ± 2.0	8.0 ± 2.6
$\dot{V}O_2$ (L·min ⁻¹)	3.99 ± 0.48	4.29 ± 0.38	4.33 ± 0.39*
RER	1.01 ± 0.03	1.00 ± 0.03	1.00 ± 0.04
Group 3			
HR (bpm)	167 ± 9	172 ± 9	173 ± 7*
Lac ⁻ (mmol·L ⁻¹)	6.6 ± 4.1	7.4 ± 3.6	7.9 ± 3.9
$\dot{V}O_2$ (L·min ⁻¹)	3.96 ± 0.52	4.12 ± 0.49	4.29 ± 0.51†
RER	0.99 ± 0.05	1.01 ± 0.04	1.00 ± 0.05
Control group			
HR (bpm)	166 ± 9	168 ± 9	167 ± 9
Lac ⁻ (mmol·L ⁻¹)	6.5 ± 2.3	6.7 ± 2.2	6.3 ± 1.8
$\dot{V}O_2$ (L·min ⁻¹)	3.98 ± 0.39	3.99 ± 0.31	3.91 ± 0.39
RER	1.00 ± 0.04	0.99 ± 0.03	0.99 ± 0.02

* $p < 0.05$ vs. before measure.

† $p < 0.01$ vs. before measure.

val training groups ($p < 0.05$) but not in the control group. Average Lac⁻ during the TT₄₀ significantly increased throughout the high-intensity interval training program in group 1 ($p < 0.05$) but not in group 2, group 3 (both $p < 0.2$), or the control group ($p > 0.05$). Mean $\dot{V}O_2$ during the TT₄₀ significantly increased over time in

TABLE 2. Peak oxygen uptake ($\dot{V}O_{2peak}$), the first and second ventilatory thresholds (VT₁ and VT₂), and anaerobic capacity (ANC) for the study groups measured throughout the 4-week high-intensity interval training program.

Measure	Before	During	After
Group 1			
$\dot{V}O_{2peak}$ (ml·kg ⁻¹ ·min ⁻¹)	65.6 ± 6.5	67.7 ± 6.8	69.5 ± 6.3*†
VT ₁ (L·min ⁻¹)	3.25 ± 0.22	3.47 ± 0.25	3.74 ± 0.28*
VT ₂ (L·min ⁻¹)	3.83 ± 0.40	4.16 ± 0.28	4.43 ± 0.22*
ANC (ml·min ⁻¹)	20.2 ± 10.9	25.1 ± 10.2	41.3 ± 20.1‡
Group 2			
$\dot{V}O_{2peak}$ (ml·kg ⁻¹ ·min ⁻¹)	66.3 ± 4.3	68.3 ± 4.6	69.9 ± 2.7*†
VT ₁ (L·min ⁻¹)	2.99 ± 0.38	3.32 ± 0.21	3.63 ± 0.26*†
VT ₂ (L·min ⁻¹)	3.82 ± 0.42	4.15 ± 0.38	4.41 ± 0.45*
ANC (ml·min ⁻¹)	27.4 ± 7.9	31.2 ± 16.8	42.3 ± 15.8‡
Group 3			
$\dot{V}O_{2peak}$ (ml·kg ⁻¹ ·min ⁻¹)	63.7 ± 3.8	66.2 ± 4.9	66.1 ± 5.8
VT ₁ (L·min ⁻¹)	3.16 ± 0.59	3.33 ± 0.57	3.69 ± 0.52*§
VT ₂ (L·min ⁻¹)	3.82 ± 0.44	3.98 ± 0.45	4.15 ± 0.57*
ANC (ml·min ⁻¹)	24.7 ± 8.3	23.1 ± 7.7	43.3 ± 19.5*
Control group			
$\dot{V}O_{2peak}$ (ml·kg ⁻¹ ·min ⁻¹)	68.3 ± 4.6	68.3 ± 4.2	68.8 ± 4.1
VT ₁ (L·min ⁻¹)	3.16 ± 0.39	3.18 ± 0.34	3.24 ± 0.41
VT ₂ (L·min ⁻¹)	3.91 ± 0.31	3.91 ± 0.33	3.90 ± 0.37
ANC (ml·min ⁻¹)	29.3 ± 11.3	26.8 ± 8.7	32.2 ± 10.0

* $p < 0.01$ vs. before measure.

† Change in $p < 0.01$ vs. control group.

‡ $p < 0.05$ vs. before PRE measure.

§ Change in $p < 0.05$ vs. control group.

TABLE 4. Relationship between changes in 40-km time trial performance and changes in physiological variables.*

	$\dot{V}O_{2peak}$	VT ₁	VT ₂	ANC	PV	TT ₄₀ $\dot{V}O_2$	TT ₄₀ HR	TT ₄₀ Lac ⁻
TT ₄₀	0.41†	0.34†	0.42‡	0.40†	0.01	0.53‡	0.42‡	0.43‡
$\dot{V}O_{2peak}$		0.41†	0.42‡	0.21	0.20	0.32	0.12	0.44‡
VT ₁			0.70‡	0.20	-0.11	0.36†	0.09	0.27
VT ₂				0.45‡	0.12	0.27	0.28	0.40†
ANC					0.09	0.18	0.22	0.34†
PV						-0.23	-0.08	0.11
TT ₄₀ $\dot{V}O_2$							0.22	0.12
TT ₄₀ HR								0.61†

* $\dot{V}O_{2peak}$ = peak oxygen uptake; VT₁ = first ventilatory threshold; VT₂ = second ventilatory threshold; ANC = anaerobic capacity; PV = plasma volume; TT₄₀ $\dot{V}O_2$ = oxygen uptake during 40-km time trial; TT₄₀ HR = heart rate during 40-km time trial; TT₄₀ Lac⁻ = blood lactate during 40-km time trial.

† Correlation significant at the $p < 0.05$ level.

‡ Correlation significant at the $p < 0.01$ level.

groups 1 and 2 (both $p < 0.05$) and group 3 ($p < 0.01$) but not in the control group. The RER measured during the TT₄₀ remained unchanged for all groups, and there were no significant differences among the groups in the mean TT₄₀ HR, Lac⁻, $\dot{V}O_2$, and RER measured throughout the high-intensity interval training program. The changes in TT₄₀ performance and measured physiological variables are presented in Table 4.

DISCUSSION

The present study has shown that high-intensity interval training, in various forms, results in significant increases in VT₁, VT₂, and ANC in already well-trained athletes (Table 2). These increases were significantly associated with improved TT₄₀ performance (Table 4). However, high-intensity interval training had no significant influence on PV (Table 1). Improvements in TT₄₀ performance were also accompanied by, and related to, significant increases in $\dot{V}O_2$ (+3.5–8.5%) and HR (+3.5%) during the TT₄₀ (Tables 3 and 4). The higher HR obtained during the TT₄₀ was also associated with an ability to tolerate more lactate ($r = 0.61$; $p < 0.01$; Table 4). Lac⁻ measured during the TT₄₀ increased by 31% in group 1 ($p < 0.05$) and showed trends toward statistically increasing in group 2 (+13%) and group 3 (+16%; both $p < 0.2$). Thus, it is likely that the improvements found in TT₄₀ performance (19) following high-intensity interval training were in part accomplished through increases in oxygen uptake, an enhanced glycogenolytic flux, and an improved ability to buffer H⁺ ions (34).

Both VT₁ and VT₂ were significantly ($p < 0.01$) increased in group 1 (+15–16%), group 2 (+17–24%), and group 3 (+9–18%), but not in the control group (0–3%; Table 2). The significant increases in VT₁ and VT₂ are similar to those we have previously shown in well-trained cyclists (+15–22%) following 4 high-intensity interval training sessions during 2 weeks (19). Indeed, improvement in the VT following an augmented training load in well-trained athletes is a consistent finding (1, 12, 15). Assuming the VT to be an indicator of the Lac⁻ threshold (29), this suggests that improvements in endurance performance with well-trained athletes following high-intensity interval training are achieved through peripheral skeletal muscle adaptations that enhance fat relative to carbohydrate oxidation at the same absolute exercise intensity (33). However, the more invasive studies to date with well-trained athletes are inconsistent with respect to this issue. High-intensity interval training in highly

trained cyclists has been shown to have no effect on 3-hydroxyacyl coenzyme A dehydrogenase and citrate synthase activity (34). In contrast, an earlier study in highly trained runners has shown that higher training intensities were associated with an increase in the oxidative enzyme citrate synthase (+18%; $p < 0.05$) (28). Clearly, further research that examines the effect of high-intensity interval training on muscle enzymatic adaptations in already well-trained athletes is required.

Estimation of ANC using the accumulated oxygen deficit method (22) has certain limitations (21), but it can provide a valid measure of anaerobic energy release (23). The measures of ANC found with the present subjects before high-intensity interval training (Table 2) are lower than those recently reported by Gardner et al. (6) for highly trained cyclists, but our posttraining values are comparable (6). In the present study, ANC was significantly ($p < 0.05$) increased in group 1 (+104%), group 2 (+54%), and group 3 (+75%), but not in the control group (+9%; Table 2). The present improvements in ANC are larger than those previously reported following 6 weeks of sprint training (+28%) (30), and there are at least 3 explanations for this. First, subjects in the present study were in their “off-season” and training was low intensity in nature. Second, the maximal exercise trial used to determine oxygen deficit was completed at a lower intensity (i.e., P_{max}) than that traditionally used for maximal accumulated oxygen deficit calculations (6). This longer trial would have resulted in a greater total $\dot{V}O_2$ and therefore a lower oxygen deficit. Third, establishment of the accumulated oxygen deficit regression equation using progressive exercise intensities has been recently shown to result in a lower ANC estimate (6). Nevertheless, the use of this method was convenient (6) and provided us with insight into the influence of the various forms of high-intensity interval training on ANC and its relationship to TT₄₀ performance. Indeed, improvement in TT₄₀ performance and increases in the associated Lac⁻ response were modestly related to changes in ANC ($r = 0.34$ – 0.43 ; $p < 0.05$). This supports the notion that an improved glycogenolytic capacity contributes to the enhanced endurance performance following high-intensity interval training in already well-trained athletes (2). This finding also supports the work of Weston et al. (34), who showed that an increased skeletal muscle buffering capacity was related to an improved TT₄₀ performance following high-intensity interval training in well-trained cyclists.

Another possible reason for the relatively low pre-training levels of ANC found in the present study was that we did not adjust for the influence of oxygen stores in our calculation, which can contribute up to 10% error (21). Although the influence of high-intensity interval training on myoglobin levels in already well-trained athletes is unknown, increases have been reported following endurance training in animals (24) and in well-trained runners during the 24-hour period following an endurance event (7). It is possible that some of the present improvements found in ANC and endurance performance following high-intensity interval training could have been partially due to increases in myoglobin levels. Notwithstanding the limitations to our ANC method, the present study is, to our knowledge, the first to show associated increases in ANC, Lac^- , and TT_{40} following high-intensity interval training in already well-trained athletes. Moreover, the increases in VT_1 , VT_2 , and ANC suggest that more than one mechanism may have contributed to the improved endurance performances following high-intensity interval training.

Another important finding in the present study was that resting PV remained unchanged in response to the high-intensity interval training programs (Table 1). Expansion of PV has long been regarded as the most important event responsible for the improved physical work capacity following short-term exercise training in untrained individuals (27). However, highly trained athletes already possess elevated levels of PV, so further improvements might not occur with an enhanced training load (32). Despite this, Richardson et al. (26) showed that both PV and stroke volume increase by 4% each in well-trained runners following 2 consecutive days of high-intensity interval training. However, these runners ($\dot{V}\text{O}_{2\text{max}} = 60.7 \pm 3.6 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$), who were training $53.8 \pm 5.6 \text{ km}\cdot\text{wk}^{-1}$, would not be considered elite, and the authors did not assess whether endurance performance was improved following the high-intensity interval training sessions (26). The present study found no evidence of an enhanced PV (Table 1), and there was no relationship between changes in PV and changes in TT_{40} performance (Table 4). This suggests that well-trained endurance athletes are at or near the limits of their diastolic reserve capacity and that further PV expansion in response to training is unlikely to further enhance cardiac function (31, 32). In those studies that have shown a significant PV expansion (14, 25, 26), blood was sampled within the 24-hour period following the high-intensity exercise sessions. In the present study, blood was sampled on the day of the athletes' endurance performance assessment, which usually followed 2 or 3 recovery days. Thus, although it is possible that high-intensity interval training may induce a transient expansion of PV in the days immediately following a training session (14, 25, 26), our findings suggest that it returns to baseline within 3–4 days and therefore is not likely to be responsible for the further improvements in endurance performance that occur following high-intensity interval training in already well-trained athletes.

In conclusion, the improvements in TT_{40} performance we have shown previously (19) were accompanied by significant increases in $\dot{V}\text{O}_{2\text{peak}}$, VT_1 , VT_2 , and ANC in all high-intensity interval training groups but not in low-intensity controls. Moreover, standard markers of PV were not significantly altered throughout the training programs. More invasive studies are required to examine the

central and peripheral adaptations that may occur following different high-intensity interval training programs to explain improvements in performance in the already well-trained athlete.

PRACTICAL APPLICATIONS

Although the adaptations to various forms of endurance training techniques in untrained individuals are well known, less is understood about these adaptations when individuals are already well trained. The present study examined the influence of 3 different high-intensity interval training programs in already well-trained cyclists ($\dot{V}\text{O}_{2\text{peak}} = 64.5 \pm 5.2 \text{ ml}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$) on changes in TT_{40} performance, $\dot{V}\text{O}_{2\text{peak}}$, the VTs, ANC, and resting levels of PV. We found that the significant improvements that occurred in TT_{40} performance were accompanied by significant increases in $\dot{V}\text{O}_{2\text{peak}}$, the VTs, and ANC, following various forms of high-intensity interval training (repeated supramaximal sprints and repeated prolonged [2–3 minutes] peak aerobic power training). Standard markers of resting PV were not significantly altered throughout any of the training programs. A closer look at the data supports the principle of specificity, suggesting that the implementation of a repeated supramaximal sprint interval training program improves the capacity for adenosine triphosphate production through glycolytic means (supported by greater changes in Lac^- levels during the TT_{40} and increases in ANC), whereas the implementation of a more prolonged peak aerobic power interval training program improves the capacity of adenosine triphosphate production via oxidative phosphorytic means (supported by greater increases in $\dot{V}\text{O}_{2\text{peak}}$ and the VTs). Thus, more than one adaptation may be responsible for the improved performances witnessed following high-intensity interval training in well-trained endurance athletes. Although future research is clearly needed in this area to support these claims, the conditioning coach may wish to consider implementing a variety of exercise intensities, durations, and recoveries into the training programs of their elite endurance athletes, depending on the physical demands of their sport.

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